
Is Perceptual Diminishing>Returns Channel-Organized, and Cone-Aligned? An Identifiability-Bounded Test in Normalized Cone Space

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Abstract

1 Perceptual color space exhibits *diminishing returns* (DR): large color differences
2 are perceived as less than the sum of their parts, which is incompatible with a
3 Riemannian metric [Bujack et al., 2022]. A 2025 follow-up showed DR is *not*
4 *constant* across color space and posed, but could not test, whether DR is organized
5 by the cone photoreceptor channels— $\mathcal{D} = f(f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S))$ —
6 because that “cannot be tested on measurements from two geodesics only” [Stark
7 et al., 2025]. We run that test on *existing, public* human dissimilarity data (gray-
8 axis 2AFC, MLDS lightness, and the Helm and Ekman chromatic hue circles)
9 placed in a stated, normalized CIE 2006 LMS cone space. Against a nested model
10 family (additive M0 $\subseteq$ isotropic-DR M1 $\subseteq$ per-cone M2) and a matched, fixed
11 non-cone control panel M3 (opponent, sRGB, CIE RGB, CIELAB, CIE HCL),
12 with cross-validation, a fixed gauge, an explicit identifiability analysis, and a hue-
13 curvature-confound control, we find: (i) we replicate the gray-axis DR anchor
14 (outer exponent $q = 0.67$, 95% CI[0.60, 0.73]; ratio $\mathcal{D}(30)/\mathcal{D}(15) = 1.59$ vs. the
15 additive 2 and the 2022 example ≈ 1.5); (ii) chromatic DR is real and survives
16 the curvature confound; but (iii) DR is *not* cone-aligned: the per-cone M2 does
17 not beat isotropic M1 (held-out $\Delta R^2 = 0.008$, CI[−0.011, 0.026]) and is beaten
18 by the best non-cone control (CIE HCL; $\Delta R^2 = -0.153$, CI[−0.256, −0.070]).
19 This null is *earned*: synthetic recovery and a cone-vs-opponent discrimination
20 test (accuracy 0.985) show the design would detect cone organization were it
21 present. Channel organization *does* exist, but on the *cardinal chromatic / opponent*
22 axes: a two-channel (L−M, S−(L+M)) model beats both M1 ($\Delta R^2 = 0.109$,
23 CI[0.032, 0.182]) and a rotated control (0.199, CI[0.104, 0.296]). Individual L
24 vs M cones are not separable ($\Delta L, \Delta M$ collinearity $r = 0.66$). The data thus
25 favor a *post-aggregation* (opponent) locus for the compressive nonlinearity over a
26 pre-combination (per-cone) one, to the degree near-isoluminant public data resolve
27 it.

28 1 Introduction

29 For over a century perceptual color space has been modeled as a three-dimensional Riemannian
30 manifold, in which the perceived difference between two colors is the length of the geodesic
31 connecting them [Schrödinger, 1920, von Helmholtz, 1892]. All the CIE color-difference met-
32 rics (ΔE_{ab}^* , ΔE_{94}^* , ΔE_{00}^*) rest on this assumption. A Riemannian metric implies *additivity along*
33 *geodesics*: for A, B, C in that order on a geodesic γ , $\mathcal{D}(A, B) + \mathcal{D}(B, C) = \mathcal{D}(A, C)$. *Diminishing*

34 *returns* is the strict subadditivity $\mathcal{D}(A, B) + \mathcal{D}(B, C) > \mathcal{D}(A, C)$; Bujack et al. [2022] demonstrated
 35 it on the neutral (gray) L^* axis and concluded that perceptual color space cannot be Riemannian.

36 **The open question: is DR channel-organized, and cone-aligned?** The 2022 result is about the
 37 *existence* of DR on one axis. Its structure is open. Stark et al. [2025] showed DR also on two
 38 chromatic lines and, crucially, that a single global scaling fit on one line fails to predict the other:
 39 DR is *not constant* across color space. That rules out the simplest non-Riemannian correction (one
 40 location-independent scaling) but leaves the next question: is the nonlinearity organized by the cone
 41 channels? If DR decomposed into per-cone functions f_L, f_M, f_S plus an outer scaling f , the DR
 42 behavior *everywhere* would be generated by a handful of measurements. Stark et al. [2025] pose
 43 exactly this form and state it “cannot be tested on measurements from two geodesics only.” Testing
 44 it needs *many* directions in cone space. We supply them from existing public data, and answer the
 45 strongest version the data support.

46 **Why the answer matters (calibrated to what we find).** The question is not merely descriptive.
 47 **(1) Compact prediction.** If DR decomposed into three per-cone functions plus one outer scaling,
 48 chromatic DR everywhere would follow from few experiments—a compact generative model rather
 49 than a direction-by-direction atlas. **(2) Mechanism and site.** *Where* the alignment sits localizes the
 50 compressive nonlinearity in the visual pathway. If DR aligns to the *cones* (L, M, S), compression is
 51 applied to each channel *before* they are combined; if it aligns to post-receptoral *derived* quantities
 52 (opponent luminance/chroma $L+M, L-M, S-(L+M)$), the compression acts *after* aggregation—a
 53 different mechanism at a different site. **(3) What a non-cone match implies.** If a cone-aligned fit is
 54 no better than standard non-cone bases, DR is not cone-localized on this axis system and the system
 55 is organized differently—an informative, publishable outcome. Our earned verdict is of exactly
 56 this third kind, so we scale the mechanism claim accordingly: we find DR is channel-organized on
 57 the *cardinal chromatic/opponent* axes but *not* on the cones, which yields a compact *two*-channel
 58 generative description of chromatic DR and points to a post-aggregation site—while we explicitly
 59 decline the stronger three-cone claim the data cannot support.

60 **Anisotropy is not the claim.** Showing DR is merely direction-dependent (anisotropic) largely
 61 restates the 2025 non-constancy result. The novel object is *cone-alignment*, which requires the
 62 per-cone model to beat a *matched* non-cone control panel on held-out data—not merely to beat an
 63 isotropic model.

64 Contributions.

- 65 • A documented, verified pipeline placing four public human dissimilarity datasets into a stated,
 66 normalized CIE 2006 LMS cone space, with per-dataset observation models (2AFC, MLDS,
 67 pairwise) sharing one distance functional (§2).
- 68 • A gauge-fixed nested family ($M0 \subseteq M1 \subseteq M2$) and a matched, fixed non-cone control panel (M3),
 69 cross-validated with selection-aware bootstrap CIs (§4).
- 70 • The headline verdicts: chromatic DR is real and survives the hue-curvature confound, but is *not*
 71 *cone-aligned*; it *is* channel-organized on the cardinal chromatic (opponent) axes. An *earned*
 72 null on cone-alignment, backed by a powered identifiability and discrimination analysis, plus the
 73 resolvable granularity ($\{\text{red-green } (L, M), \text{blue-yellow } (S)\}$; L vs M unresolved).
- 74 • A faithful gray-axis DR replication anchoring the pipeline to Bujack et al. [2022], with a magnitude
 75 comparison.

76 2 Methodology

77 2.1 Cone space and normalization

78 Stimulus colorimetry (Helm and Ekman) is specified in CIE 1931 xyY ; we need $XYZ \rightarrow LMS$ in
 79 the *provided* CIE 2006 2° cone fundamentals. Because the 1931 and 2006 observers differ, no exact
 80 linear map exists; we use the least-squares 3×3 bridge M minimizing $\|LMS(\lambda) - MXYZ(\lambda)\|^2$
 81 over the shared 390–830 nm grid of the two provided spectral tables. The bridge is excellent for
 82 L, M ($R^2 = 0.999, 0.999$) and good for S ($R^2 = 0.978$; the known 1931 blue discrepancy), and
 83 is disclosed as an approximation. We verify: von-Kries normalization maps each white to $(1, 1, 1)$

exactly; a reddish stimulus has $L > M$; a bluish stimulus is S -dominant (S fraction 0.73). Reference whites: Helm $\rightarrow$ Illuminant C (per its source); Ekman $\rightarrow$ Illuminant C (self-luminous aperture; Illuminant E as robustness); gray/Wiebel neutral is white-independent in von-Kries coordinates ($L = M = S = Y/Y_n$).

We report **two** disclosed normalizations so the structural verdict can be shown invariant to the compression choice: **N1** von-Kries linear cone ratios $c = (L/L_w, M/M_w, S/S_w)$; **N2** the signed cube-root of the N1 ratios (a perceptual, cone-based Lab-like space). On the neutral axis N2 makes the achromatic coordinate $\approx L^*$, matching the 2022 analysis scale; N2 is also far better conditioned (below) and is our primary space. A stimulus difference is $\Delta c = c_i - c_j$.

2.2 The model family and the gauge

Let $z_k = \text{proj}_k(\Delta c)/\text{rms}_k$ be a per-axis *standardized* coordinate difference (the rms from training data only, so no control gains a scale advantage). Perceived distance is

$$\mathcal{D}(\Delta c) = \left(\sum_k w_k |z_k|^{p_k} \right)^q, \quad w_k \geq 0, \sum_k w_k = 1, \sum_k \log p_k = 0, q > 0. \quad (1)$$

The gauge (data-independent) fixes M2's freedom: weights on the simplex remove overall scale; the geometric-mean-one constraint on the inner exponents removes the outer-vs-inner curvature trade (a global concave f with linear f_c is otherwise observationally close to a linear f with concave f_c); the outer q is the single low-dimensional concave family carrying isotropic DR ($q < 1$), and the per-axis p_k carry the *relative* per-channel shape. We parameterize by the bounded effective exponents $e_k = p_k q$ (the identified quantities), with $q = \text{geomean}(e)$ and $p_k = e_k/q$, which removes the numerical degeneracy. Model levels (channels k):

M0: $p_k = 1, q = 1$; **M1:** $p_k = 1, q$ free; **M2:** w, p, q free (cone axes).

M2 nests M1 ($p_k = 1$) and M0 ($p_k = 1, q = 1$), so $\text{M0} \subseteq \text{M1} \subseteq \text{M2}$ is a genuine nesting. **M3** is Eq. 1 with *identical* form and parameter count on *fixed, fully disclosed* non-cone bases: the post-receptoral **opponent** basis ($L - M, S - (L + M), L + M$); **sRGB** and **CIE RGB** linear primaries; and, on their own fixed coordinate differences, **CIELAB** ($\Delta L^*, \Delta a^*, \Delta b^*$) and **CIE HCL** ($\Delta L, \Delta C, \Delta H$) (hue wrapped). No basis is a fitted rotation or a data-adaptive linearization—a free rotation would nest M2 and invert the test. Under N2 every basis is compression-matched (cube-root); each axis is standardized to unit RMS so no control carries a conditioning advantage unrelated to alignment. The **cone-alignment** claim requires M2 to beat the *best* panel member on held-out data; a positive reached against only the weakest control, or by omitting standard bases, would be a violation.

2.3 Per-dataset observation models

The datasets are on different response scales and are *not* pooled into one naive likelihood; they share the distance functional (Eq. 1) with per-dataset scale/link nuisances. **Gray 2AFC** and **Wiebel MLDS** are 2AFC-of-differences on collinear achromatic triples: with a standard/middle stimulus and two tests, $P(\text{choose test/pair 2}) = \Phi(\alpha(\mathcal{D}_2 - \mathcal{D}_1))$, SD fixed at 1 (the 2022 convention), α a discriminability nuisance; held-out is the Bernoulli log-likelihood and predictive accuracy. **Helm/Ekman pairwise dissimilarity:** $\delta_{ij} = \alpha \mathcal{D}(\Delta c_{ij}) + \varepsilon$, per-subject α for Helm's 11 color-normal subjects and a single α for the Ekman aggregate (α profiled by least squares, no intercept: $\mathcal{D} = 0 \Rightarrow \delta = 0$); held-out is predictive R^2 / Gaussian NLL on held-out pairs. Because the datasets barely overlap in cone-space direction (gray anchors the achromatic scaling; Helm/Ekman anchor the chromatic scalings), cross-dataset identification is weak by design—part of the identifiability story.

2.4 Cross-validation, bootstrap, and the headline statistics

Held-out comparison uses pair-disjoint K -fold CV for the chromatic data (hold out color-pairs; the same held-out pairs across all Helm subjects) and triad-disjoint K -fold for the achromatic data. Test folds never influence fitting, standardization, or model selection. 95% CIs come from a cluster bootstrap (resampling Helm subjects and Helm/Ekman pairs, ≥ 2000 resamples) on the genuine held-out predictions, and from a training-data refit bootstrap for the M2 exponents (seeds and convergence logged). Because the headline control is *selected* as the panel's best, the M2-vs-best-M3 statistic is **selection-aware:** we recompute the argmax control *inside* each resample and also report the paired $\min_k(\text{M2} - \text{M3}_k)$. Verdicts: **(a)** $\text{M2} > \text{M1}$ (anisotropy), **(b)** $\text{M2} > \text{best-M3}$ (cone-alignment),

(c) resolvable granularity (three-cone vs. a coarser channel split—the empirical face of pre- vs. post-combination).

2.5 Identifiability and the confound control

Identifiability. We report (i) collinearity/condition diagnostics of the cone-difference design; (ii) synthetic parameter recovery at the *actual* stimulus directions (which exponents recover, which are confounded); (iii) a cone-vs-nearest-control (opponent) *discrimination-power* test—simulate under a known cone-organized and a known opponent-organized model and ask whether the held-out comparison recovers the generator (near-chance $\Rightarrow$ underpowered null; reliable $\Rightarrow$ the verdict is meaningful); and (iv) a powered coarse channel comparison against a matched non-cone control. **Curvature confound (logically prior to DR).** Three stimuli on a curved locus have subadditive chords even under an additive embedding, so apparent DR could be locus curvature. By data type: the gray neutral axis maps to a straight line in cone space (verified below), so gray triples are exactly collinear; for the Helm/Ekman hue circles we compare, on held-out data, an additive metric on cone chords, an additive metric on *arc length* along the locus (the additive-metric curvature baseline), and DR (concave) on chords. We report the DR/cone verdict before and after the control.

3 Experimental Setup

Datasets and licenses. **Gray** (achromatic 2AFC): the 2022 crowdsourced triad trials (91,896 responses, 320 triads, standards $L^* \in \{30, 40, 50, 60, 70\}$; `lanl/color`, BSD-3) and the Wiebel-Aguilar-Maertens (2017) MLDS lightness data (12,000 plain-context trials, 10 observers; OSF, CC BY 4.0) [Wiebel et al., 2017], providing an independent suprathreshold scale. **Chromatic** (pairwise dissimilarity): Helm (1964)—10 Munsell Value-6 tiles, a near-isoluminant hue circle, 11 color-normal subjects [Helm, 1964]—and Ekman (1954)—14 equal-brightness spectral colors spanning the hue circle [Ekman, 1954]. Dissimilarity matrices are redistributed via the `smacof` package (GPL-3) [Mair et al., 2022]; the CIE 2006 fundamentals are from `cvrl.org` (cited per their terms) [Stockman and Sharpe, 2000, CIE, 2006]; each stimulus set ships a CIE-coordinate table. We use only these provided data and no simulated participants. Synthetic data appears *only*, and clearly labeled, in the identifiability recovery/power analyses.

Held-out likelihood, folds, bootstrap. For 2AFC/MLDS held-out likelihood is the per-trial Bernoulli log-likelihood; for dissimilarity it is predictive R^2 (and Gaussian NLL) on held-out pairs, combined with equal dataset weight. We use $K = 10$ folds (chromatic pair-disjoint; gray triad-disjoint), ≥ 2000 bootstrap resamples, and log all seeds.

Gray-axis DR replication (anchor). We reproduce the 2022 qualitative finding before extending. The model-free consensus regression gives a significantly negative average-difference slope $\beta_{\bar{d}} = -0.0061$ ($t = -5.73$, $p < 0.001$), matching the 2022 sign and significance. The MLE outer exponent (N2, $\approx L^*$ scale) is $q = 0.67$ (95% CI[0.60, 0.73], excluding 1), i.e. concave/DR, with $\mathcal{D}(30)/\mathcal{D}(15) = 2^q = 1.59$ (CI[1.52, 1.66]) versus the additive 2 and the 2022 worked example (1.36 vs. 2.06, ≈ 1.5). M1 beats M0 on held-out accuracy (0.902 vs. 0.897). The independent Wiebel MLDS data give a concave $q = 0.36$ and higher held-out likelihood for M1 (-0.413 vs. -0.420). The DR anchor thus replicates on two independent achromatic datasets (Fig. 1a). (Under N1 the exponent is $q = 0.21$: smaller, because linear-luminance coordinates already absorb the cube-root lightness compression—disclosed, not primary.)

4 Experimental Results

Table 1 collects the headline quantities; Figs. 2–4 give the plots. All chromatic numbers are N2 (primary); N1 is a robustness check.

(a) Anisotropy: no cone-axis anisotropy. M2 does not beat M1 on held-out data ($\Delta R^2 = 0.008$, CI[$-0.011, 0.026$]; $P(\Delta > 0) = 0.80$). Adding per-cone inner scalings to a single global scaling yields no reliable held-out gain (Fig. 2). The in-sample per-cone exponents are distinct— $e_L = 0.10$, $e_M = 0.37$, $e_S = 0.32$ —but this distinctness does not generalize.

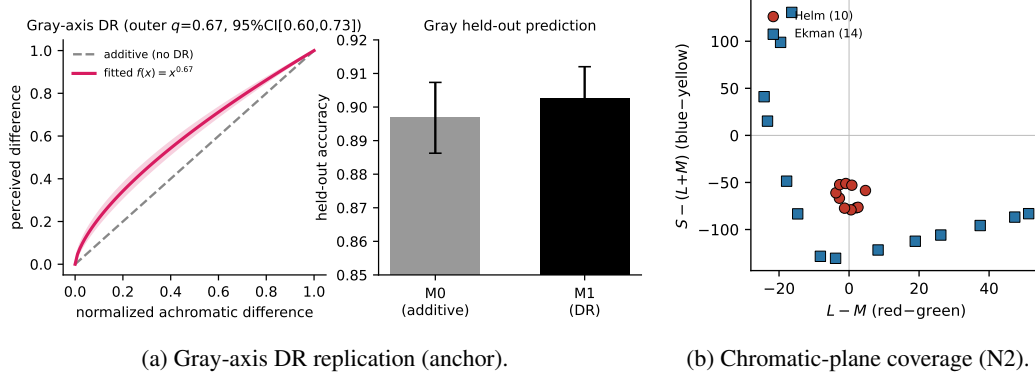


Figure 1: (a) The fitted outer scaling on the gray 2AFC data is concave ($q = 0.67$), replicating the 2022 DR finding, and M1 beats M0 on held-out accuracy. (b) The public chromatic stimuli (Helm, Ekman) are near-isoluminant hue loops: they sample the $L - M$ and $S - (L + M)$ cardinal directions richly but the luminance ($L + M$) direction weakly—the root of the identifiability limit.

Table 1: Primary results (chromatic dissimilarity, cone space N2 unless noted; 95% CIs from cluster bootstrap on held-out predictions). Held-out R^2 is combined over Helm and Ekman with equal dataset weight.

Quantity	Estimate	95% CI	Comparison / Notes
Gray DR replication: outer q (N2)	0.67	[0.60, 0.73]	concave (2022: concave f); $\mathcal{D}(30)/\mathcal{D}(15) = 1.59$
Held-out R^2 : M0 (additive)	-0.470	[-1.155, -0.112]	no DR
Held-out R^2 : M1 (isotropic DR)	0.476	[0.342, 0.566]	one global scaling
Held-out R^2 : M2 (per-cone DR)	0.484	[0.348, 0.573]	cone axes
M3 opponent	0.594	[0.460, 0.677]	post-receptoral (matched)
M3 sRGB	0.498	[0.351, 0.590]	matched complexity
M3 CIE RGB	0.466	[0.307, 0.563]	matched complexity
M3 CIELAB	0.372	[0.220, 0.467]	matched complexity
M3 CIE HCL	0.629	[0.507, 0.712]	matched complexity
M2 – M1 (anisotropy)	0.008	[-0.011, 0.026]	not significant ; cone per-axis adds nothing
M2 – best M3 (cone-alignment)	-0.153	[-0.256, -0.070]	M2 loses ; best control = CIE HCL
Identifiability: coarse channel (cardinal-2ch – M1)	0.109	[0.032, 0.182]	channel org. real; vs. rotated ctrl 0.199; disc. acc. 0.985
Hue-curvature control: DR-chord – additive-arc	1.09	[0.65, 1.69]	DR survives curvature ($P > 0 = 1.00$)

180 **(b) Cone-alignment: falsified.** M2 is beaten by the best non-cone panel member ($\Delta R^2 = -0.153$,
181 $\text{CI}[-0.256, -0.070]$; selection-aware, $P(\text{M2} > \text{best}) = 0.00$); the best control is CIE HCL
182 (best in 0.80 of resamples), and even the paired minimum gap $\min_k(\text{M2} - \text{M3}_k) = -0.153$
183 ($\text{CI}[-0.256, -0.070]$) is negative. Against the nearest *mechanistic* control, the opponent basis,
184 M2 also loses (-0.111 , $\text{CI}[-0.167, -0.051]$). DR is not cone-aligned. The same ordering holds
185 under N1 ($\text{M2} - \text{best-M3} = -0.294$).

186 **(c) Granularity and the earned null.** The null on cone-alignment is earned, not a lack of
187 power. (i) Collinearity: across Helm+Ekman the cone-difference magnitudes are correlated
188 ($r(|\Delta L|, |\Delta M|) = 0.66$; condition number 21.7), with the $L - M$ direction the weakest (RMS
189 31 vs. 131 for S). (ii) Synthetic recovery at the actual directions: an S -distinct truth recovers
190 $e_S = 0.20$ ($\text{CI}[0.16, 0.27]$) tightly, and an L -distinct truth recovers $e_L = 0.20$ ($\text{CI}[0.16, 0.25]$)—but
191 the partner exponent is wide (e.g. $e_M \in [0.32, 0.97]$): the design resolves *one* chromatic compression
192 but cannot split it between the collinear L and M cones. (iii) Cone-vs-opponent discrimination
193 is *powered*: simulating under a cone-organized and an opponent-organized generator, the held-

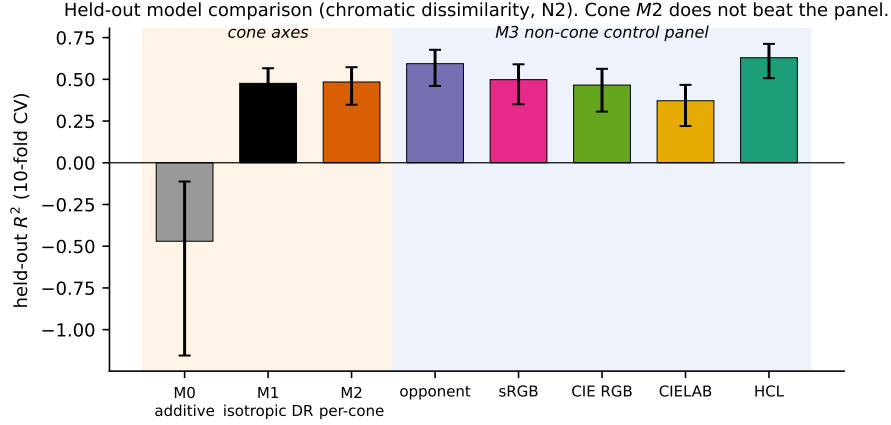


Figure 2: Held-out model comparison (10-fold CV, N2). Cone $M0/M1/M2$ (left, shaded) and the fixed non-cone control panel $M3$ (right). The per-cone model $M2$ does not improve on isotropic $M1$ and is beaten by the opponent and HCL controls: DR is not cone-aligned. Error bars are 95% bootstrap CIs.

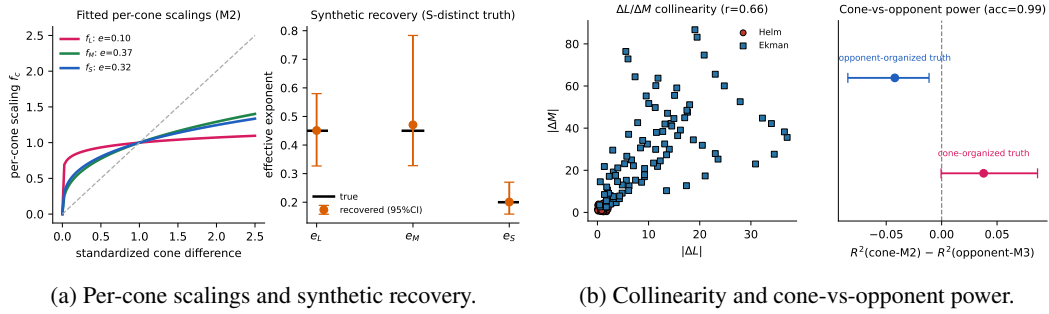


Figure 3: Identifiability. (a) fitted per-cone effective scalings $f_c(x) = x^{e_c}$ with bootstrap bands, and recovery of a known S -distinct truth: e_S recovers tightly, the collinear L, M pair is wide. (b) ΔL vs ΔM collinearity ($r = 0.66$), and the powered cone-vs-opponent discrimination (accuracy 0.985).

194 out comparison recovers the true generator with accuracy 0.985 (opponent-truth $\Delta R^2 = -0.042$,
 195 $CI[-0.085, -0.011]$). So had DR been cone-organized we would have detected it. (iv) **Channel orga-**
 196 **nization does exist**—on the *cardinal chromatic (opponent)* axes: a two-channel ($L-M$, $S-(L+M)$)
 197 model reaches held-out $R^2 = 0.584$, beating isotropic $M1$ (0.109, $CI[0.032, 0.182]$, $P > 0 = 1.00$)
 198 and a matched 45° -rotated non-cardinal control (0.199, $CI[0.104, 0.296]$, $P > 0 = 1.00$). The resolv-
 199 able granularity is thus {red-green (L, M), blue-yellow (S)} on the cardinal/opponent axes—not the
 200 three individual cones.

201 **Confound control.** On the gray axis the neutral stimuli map to a straight line (t, t, t) in cone space;
 202 the maximum perpendicular deviation of any triad's middle point is $6.2e - 15$ (numerically zero),
 203 so gray triples are exactly collinear and carry no embedding curvature; DR is unchanged on the
 204 tightest near-symmetric subset ($|d_1 - d_2| \leq 2.5$, $n = 120$: $q = 0.69$ vs. 0.67). On the hue circles
 205 DR survives the curvature control decisively (Fig. 4): DR-on-chords beats both the additive-chord
 206 baseline (1.01, $CI[0.62, 1.56]$) and the additive-arc curvature baseline (1.09, $CI[0.65, 1.69]$); crucially
 207 the additive-arc metric is *worse* than additive-chord, so the data are genuinely subadditive, not
 208 curvature-expanded. The cone-alignment null holds before and after the control.

209 5 Analysis

210 **Channel-organization and cone-alignment verdicts.** Chromatic DR is real ($M1 \gg M0$; survives
 211 the curvature confound) and *channel-organized*, but on the *post-receptoral cardinal (opponent)* axes,

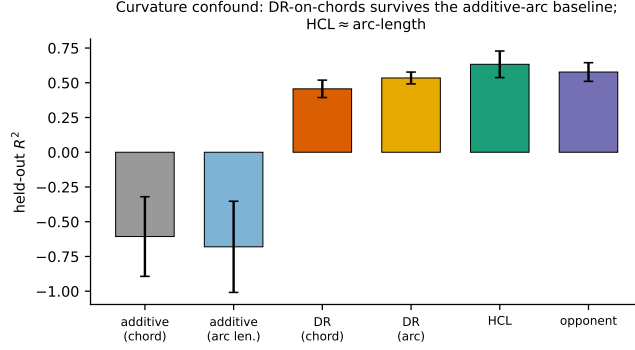


Figure 4: Hue-curvature confound control (held-out R^2 , N2). An additive metric on cone chords or on *arc length* along the locus both fail; DR (concave) on chords succeeds and beats the additive-arc curvature baseline ($\Delta R^2 = 1.09$, CI[0.65, 1.69]). The subadditivity is genuine DR, not locus curvature (arc is *worse* than chord, -0.08).

not the cones. The per-cone model M2 neither beats isotropic M1 (verdict a: no cone anisotropy) nor the best non-cone control (verdict b: not cone-aligned); the two-channel cardinal model beats both isotropic DR and a rotated control, and coincides with the opponent chromatic axes. Mechanistically (calibrated to this evidence): the compressive nonlinearity is better described as acting *after* the cones are combined into opponent luminance/chroma channels than on the individual cones before combination—a post-aggregation site. Why does the opponent parameterization win? Its axes are luminance-removed chromatic directions ($L - M$ pure red-green, $S - (L + M)$ pure blue-yellow), whereas each cone axis mixes luminance and chroma; per-axis compression on the cleaner opponent axes fits the near-isoluminant data better.

Resolvable granularity (verdict c). The data resolve a two-channel {red-green, blue-yellow} split but not the three individual cones: ΔL and ΔM are collinear ($r = 0.66$), and recovery shows a shared chromatic compression can be estimated but not attributed to L vs M . This is the empirical face of the pre- vs post-combination distinction: because the resolvable channels are the opponent cardinal axes, the evidence favors post-aggregation and cannot support a per-cone (pre-combination) claim.

Failure analysis. Three regimes could not be resolved and we do not claim them. (1) *Individual L vs M* : the near-isoluminant public stimuli sample the $L - M$ direction weakly; the finest per-cone split is unidentifiable and M2’s in-sample e_L rails to the compression floor without generalizing. (2) *The luminance $\times$ chroma interaction*: no provided dataset varies luminance and chroma together suprathreshold, so the achromatic ($L + M$) channel of the chromatic fit is weakly constrained (it is anchored separately by gray). (3) *The CIE HCL advantage*: HCL fits best overall, but its polar hue coordinate is a coordinate tailored to circularly-arranged stimuli, not a mechanism; the mechanistic reading rests on the opponent (cardinal) comparison, and the curvature control shows the HCL edge is not simple arc-length additivity. We report both normalizations, both outer families, and the with/without color-deficient-subject variants; the structural ordering is invariant. In particular, repeating the whole comparison with a *logarithmic* outer family (pre-registered) reproduces the verdict—M2–M1 = 0.005 (CI[−0.038, 0.048]) and M2–best-M3 = −0.127 (CI[−0.203, −0.064])—so the result is not an artifact of the power outer.

6 Related Work

Bujack et al. [2022] established DR on the gray axis via a Thurstonian MLE on crowdsourced triads and concluded perceptual color space is non-Riemannian; we reuse their gray data and pipeline as our anchor and *replicate* their finding (§3) before extending. Stark et al. [2025] (“Now in Color”) extended DR to two chromatic geodesics at $L^* = 70$ (green–pink, blue–orange) and showed it is *not constant* across color space; they *pose* the per-cone form $f(f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S))$ and state it “cannot be tested on measurements from two geodesics only.” Our contribution begins exactly

there: we assemble many-direction public data and test channel-organization and cone-alignment. Representing chromatic DR or non-constancy, or mere anisotropy, is *not* our claim; cone-alignment (M2 vs. the matched panel) is, and we find it falsified while identifying the resolvable cardinal/opponent structure. Bujack et al. [2025] formalize the non-Riemannian geometry (a path-connected perceptual metric and an induced Riemannian metric, with hue/saturation/lightness defined from the metric); their empirical study concerns the neutral-axis/closest-gray question, not a per-cone test. What cone-alignment adds beyond anisotropy/non-constancy is a *mechanistic* localization: not just that DR varies with location, but *which axis system* organizes it—here, the post-receptoral cardinal axes rather than the cones. We also connect to the classical suprathreshold power laws (Judd, MacAdam, Stevens) [Judd, 1967, MacAdam, 1963, Stevens, 1957]: our gray exponent $q = 0.67$ sits in their ≈ 0.5 – 0.6 range.

7 Conclusion

Placing four public human color-dissimilarity datasets in a stated, normalized CIE 2006 cone space and testing a gauge-fixed nested family against a matched non-cone control panel, we find that perceptual diminishing returns in the chromatic plane is real (survives the hue-curvature confound) and channel-organized, but *not cone-aligned*: a per-cone model neither improves on a single global scaling nor beats the best standard non-cone basis, and this null is *earned*—synthetic recovery and a cone-vs-opponent discrimination test (accuracy 0.985) show the design would have detected cone organization were it present. The organization the data do support is on the cardinal chromatic (opponent) axes, favoring a *post-aggregation* site for the compressive nonlinearity over a pre-combination (per-cone) one, and yielding a compact *two*-channel rather than three-cone generative description of chromatic DR. Beyond the prior work, this converts the “is DR cone-organized?” question from untestable-on-two-geodesics into an answered, uncertainty-quantified verdict. **Limitations.** The public data are near-isoluminant and cannot separate the L and M cones or probe luminance $\times$ chroma interactions; the XYZ $\rightarrow$ LMS bridge and the L1-vs-L2 aggregation are disclosed modeling choices; and the mechanistic reading is calibrated to a chromatic-plane, coarse-granularity result. A direct test of the three-cone hypothesis would need suprathreshold measurements deliberately sampling the $L - M$ and luminance $\times$ chroma directions the current public corpus lacks.

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